A Comparison of AI Weather Prediction and Numerical Weather Prediction Models for 1–7-Day Precipitation Forecasts

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(Manuscript received 24 May 2024, in final form 18 November 2024, accepted 10 January 2025)

ABSTRACT: Pure artificial intelligence (AI)-based weather prediction (AIWP) models have made waves within the scientific community and the media, claiming superior performance to numerical weather prediction (NWP) models. However, these models often lack impactful output variables such as precipitation. One exception is Google DeepMind's GraphCast model, which became the first mainstream AIWP model to predict precipitation, but performed only limited verification. We present an analysis of the ECMWF's Integrated Forecasting System (IFS)-initialized ($GRAP_{IFS}$) and the NCEP's Global Forecast System (GFS)-initialized ($GRAP_{GFS}$) GraphCast precipitation forecasts over the contiguous United States and compare to results from the GFS and IFS models using 1) grid-based, 2) neighborhood, and 3) object-oriented metrics verified against the fifth major global reanalysis produced by ECMWF (ERA5) and the NCEP/Environmental Modeling Center (EMC) stage IV precipitation analysis datasets. We affirmed that $GRAP_{GFS}$ and $GRAP_{IFS}$ perform better than the GFS and IFS in terms of root-mean-square error and stable equitable errors in probability space, but the GFS and IFS precipitation distributions more closely align with the ERA5 and stage IV distributions. Equitable threat score also generally favored GraphCast, particularly for lower accumulation thresholds. Fractions skill score for increasing neighborhood sizes shows greater gains for the GFS and IFS than GraphCast, suggesting the NWP models may have a better handle on intensity but struggle with the location. Object-based verification for GraphCast found positive area biases at low accumulation thresholds and large negative biases at high accumulation thresholds. $GRAP_{GFS}$ saw similar performance gains to $GRAP_{IFS}$ when compared to their NWP counterparts, but initializing with the less familiar GFS conditions appeared to lead to an increase in light precipitation.

SIGNIFICANCE STATEMENT: Pure artificial intelligence (AI)-based weather prediction (AIWP) has exploded in popularity with promises of better performance and faster run times than numerical weather prediction (NWP) models. However, less attention has been paid to their capability to predict impactful, sensible weather like precipitation, precipitation type, or specific meteorological features. We seek to address this gap by comparing precipitation forecast performance by an AI model called GraphCast to the Global Forecast System (GFS) and the Integrated Forecasting System (IFS) NWP models. While GraphCast does perform better on many verification metrics, it has some limitations for intense precipitation forecasts. In particular, it less frequently predicts intense precipitation events than the GFS or IFS. Overall, this article emphasizes the promise of AIWP while at the same time stresses the need for robust verification by domain experts.

KEYWORDS: Forecast verification/skill; Numerical weather prediction/forecasting; Artificial intelligence

1. Introduction

Pure artificial intelligence (AI)-based weather prediction (AIWP), sometimes referred to as data-driven (Pathak et al. 2022) weather modeling, has received substantial attention in the meteorological community and beyond. Over the past 2 years, several new AIWP models have been released, including Nvidia's FourCastNet (Pathak et al. 2022) and FourCastNetv2-small (Bonev et al. 2023), Huawei's Pangu-Weather (Bi et al. 2023), Google's GraphCast (Lam et al. 2023), and ECMWF's AI Forecasting System (AIFS) (Lang et al. 2024), each claiming performance comparable to or better than state-of-the-art physics-based numerical weather prediction (NWP). These claims have also been shared by the media: For example, Wired writes, "Google DeepMind's AI Weather Forecaster Handily Beats a Global Standard" (Barber 2023), while Nature states, "Google DeepMind has turned its hand to the intensive science of weather forecasting—and developed a machine-learning model that outperforms the best conventional tools, as well as other AI approaches, at the task" (Wong 2023).

Headlines and author claims of AIWP outperformance tend to be based primarily on root-mean-square error (RMSE) and/or the anomaly correlation coefficient (ACC; Pathak et al. 2022; Bi et al. 2023; Lam et al. 2023). However, a substantial amount of effort has been put in to thoroughly evaluate AIWP models compared to one another and to NWP models for a number of other features and metrics. One notable effort is WeatherBench2 (Rasp et al. 2024), which provides a framework for developers to evaluate and present their model results with a suite of verification metrics. Some AIWP developers have also verified specific meteorological features, such as temperature extremes (Lam et al. 2023), atmospheric rivers (Pathak et al. 2022; Lam et al. 2023), and tropical cyclone

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DOI: 10.1175/WAF-D-24-0081.1

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tracks (Pathak et al. 2022; Bi et al. 2023; Lam et al. 2023), with promising results. Nonetheless, some potential weaknesses of AIWP have been identified. For example, Ben Bouallègue et al. (2024) found that AIWP forecasts are too smooth or blurry, may have greater biases in longer forecast leads, and underestimate the intensity of tropical cyclones, a result echoed by DeMaria et al. (2024). Additional investigations of specific meteorological phenomena, physical processes, and extremes will be vital to further establish AIWP strengths and weaknesses and potential operational value.

Performance aside, there is some uncertainty regarding the operational utility of AIWP. First, the spatial resolutions of these models are typically limited to 0.25° or approximately 30 km. This is due to the fact that they are trained on the 0.25° fifth major global reanalysis produced by ECMWF ERA5 (Hersbach et al. 2020). For comparison, the GFS is run at 13 km, but is commonly made available to the public on coarser (such as 0.25°) grids (NWS and NCEP 2014), while the ECMWF's high-resolution forecast system (HRES) is run on a 0.10° grid and is made available on several coarser grids (ECMWF 2024). Second, temporal resolution in most cases is limited to 6-h intervals[1] (Pathak et al. 2022; Lam et al. 2023; Bonev et al. 2023). Last, the variables produced by existing AIWP models are typically limited to geopotential height, temperature, specific or relative humidity, zonal and meridional wind components, and a handful of near-surface variables (Pathak et al. 2022; Bonev et al. 2023; Bi et al. 2023). Note that this excludes key, impactful parameters such as precipitation, precipitation type, cloud cover, radiation, and wind gusts, among others.

There are some key exceptions to this last point. GraphCast was the first global AIWP model to produce precipitation and vertical velocity forecasts (Lam et al. 2023), followed more recently by the AIFS (Lang et al. 2024). Google DeepMind's MetNet-3 (Andrychowicz et al. 2023) also attempts to address issues of spatiotemporal resolution and impactful output variables but for a limited domain and short-range forecasts. AIFS and MetNet-3 model weights have not been released at the time of writing, so our focus is on GraphCast. AI-based precipitation predictions are particularly challenging due to biases in the ERA5 dataset, sparsity of precipitation observations, and non-Gaussian distribution (Lavers et al. 2022; Lam et al. 2023). Specifically, Lavers et al. (2022) found that the ERA5 24-h accumulated precipitation had an overall wet bias but often did not capture the magnitude of extreme precipitation events. Because of these known shortcomings of the ERA5 precipitation fields, the GraphCast team did not robustly evaluate the quality or attempt to further fine-tune GraphCast's precipitation forecasts but still reported lower RMSEs and lower stable equitable error in probability space (SEEPS; Rodwell et al. 2010) for GraphCast compared to the ECMWF HRES in supplemental material (Lam et al. 2023). We seek to supplement this evaluation with other grid-based verification metrics, neighborhood verification, and object-based verification to provide a robust baseline for future AIWP precipitation forecast developments.

The goals of this paper are threefold. First, we robustly evaluate GraphCast's precipitation forecasts over the contiguous United States compared to the IFS and GFS models against both the ERA5 quantitative precipitation estimates and the NCEP/Environmental Modeling (EMC) stage IV (Lin and Mitchell 2005) quantitative precipitation estimates and with multiple verification methods to establish a baseline of performance for future AIWP models to improve upon. Second, we demonstrate how which model is "better" is contextually dependent on the application, choice of verification metric, threshold, forecast lead, and ground truth dataset. Last, we include an analysis of both the "native" IFS-initialized and untested GFS-initialized versions of GraphCast, allowing for some insight into the performance of AIWP models when extended to unfamiliar initial conditions.

2. Data and methods

a. Datasets

Our first step was to produce an archive of GraphCast reforecasts, which we accomplished by initializing GraphCast with IFS ($GRAP_{IFS}$) and GraphCast with GFS ($GRAP_{GFS}$) initial conditions (i.e., F000). The operational version of GraphCast was trained on ERA5 data between 1979 and 2015 and then fine-tuned on ECMWF IFS HRES data between 2016 and 2021 (Lam et al. 2023). Note that our implementation of $GRAP_{GFS}$ has still been fine-tuned on the IFS HRES and performance may be improved in the future by fine-tuning the GraphCast model on GFS data. To limit the potential influence of verifying data that GraphCast had seen in training, our period of analysis is between 1 January 2022 and 1 October 2023. GraphCast was run twice per day with 0000 and 1200 UTC 0.25° IFS and GFS initial conditions out to 240 h for this period. The evaluated IFS and GFS forecasts were initialized from the same initial conditions.

We use two different datasets as proxies for truth in verification. The first is the ERA5 total precipitation field (Hersbach et al. 2020), which we accumulated at 6-h intervals to match GraphCast, IFS, and GFS output. ERA5 assimilates radar and rain gauge estimates with IFS precipitation forecasts (Lavers et al. 2022). The second is the NCEP/EMC stage IV 6-hourly national precipitation analysis (Lin and Mitchell 2005) dataset. The stage IV dataset is a higher resolution (4 km) than the other datasets, which we addressed by interpolating to the 0.25° latitude–longitude grid with the budget method, following Accadia et al. (2003), Wolff et al. (2014), Yan and Gallus (2016), and others. We previously noted some limitations of the ERA5 precipitation field, and the stage IV analysis has its own drawbacks. Specifically, stage IV demonstrates inconsistent quality in mountainous terrain due to limited radar coverage (Maddox et al. 2002; Kursinski and Mullen 2008). However, the point of its inclusion is not as a more accurate "ground truth." Instead, it is included as an alternative to demonstrate verification sensitivity to the choice of "truth." Since stage IV only covers the United States,

[1] An exception is Pangu-Weather (Bi et al. 2023) which provides hourly output but does not provide precipitation.

TABLE 1. Traditional contingency table.

Contingency table		
	Observed yes	Observed no
Forecast yes	Hits	False alarms
Forecast no	Misses	Correct negatives

we masked all of the forecast and observation datasets to the contiguous United States.

b. *Verification strategies*

Most studies on AIWP (Pathak et al. 2022; Bi et al. 2023; Lam et al. 2023) emphasize lower MSE or RMSE, metrics that sum differences between forecasts and observations grid point by grid point. Accordingly, we refer to these as grid-based metrics. These metrics are easily applied to virtually any variable, making them convenient for AIWP models that forecast many different variables. MSE is also the metric that most AIWP models (Pathak et al. 2022; Bonev et al. 2023; Lam et al. 2023) seek to optimize during training, although some, such as Pangu-Weather (Bi et al. 2023), use mean absolute error (MAE) instead. Other simple measures that we calculate in this section include distribution quantiles and frequencies as well as spatial bias.

More complex than MSE and RMSE, but still versions of grid-based verification are contingency table statistics, such as SEEPS (Rodwell et al. 2010), threat score [TS; also known as critical success index (CSI); Donaldson et al. 1975], and equitable TS (ETS; Gilbert 1884). These metrics compare binarized forecast and observation fields based on different thresholds and are particularly valuable when evaluating non-Gaussian distributed variables for extreme events (Rodwell et al. 2010; Rasp et al. 2024). Most commonly, and in the case of TS and ETS, these metrics use a 2×2 contingency matrix based on whether or not a forecast at a grid point exceeds a given threshold, as shown in Table 1. Thus, TS and ETS incorporate information on hits, misses, and false alarms, unlike simpler metrics like accuracy. Both TS [Eq. (1)] and ETS [Eqs. (2) and (3)] are similar to the F score Eq. (4), the harmonic mean of precision (also known as the success ratio; Brooks et al. 2024), and recall (also known as the probability of detection; Brooks et al. 2024). By applying a series of different thresholds, ETS can provide more verification granularity and highlight threshold ranges where models over- or underperform.

SEEPS uses a more complex 3×3 contingency matrix by dividing precipitation forecasts and observations into climatologically defined "dry," "light," and "heavy" at each location and uses the climatological probability of each of these categories to assign scores. Thresholds (except the dry threshold) in this case are not user defined. SEEPS is generally considered a more suitable metric for precipitation evaluation than MSE or RMSE as more intense precipitation is weighted higher, discouraging "hedging" (Rodwell et al. 2010), but does not provide as much granularity on performance as ETS evaluated at a spectrum of thresholds:

$$\mathrm{TS} = \frac{\mathrm{hits}}{\mathrm{hits} + \mathrm{misses} + \mathrm{false\ alarms}}, \tag{1}$$

$$\mathrm{ETS} = \frac{\mathrm{hits} - E}{\mathrm{hits} + \mathrm{misses} + \mathrm{false\ alarms} - E}, \tag{2}$$

where E is the number of hits expected purely by chance:

$$E = \frac{(\mathrm{hits} + \mathrm{misses}) \times (\mathrm{hits} + \mathrm{false\ alarms})}{\mathrm{hits} + \mathrm{misses} + \mathrm{false\ alarms} + \mathrm{correct\ negatives}}, \tag{3}$$

$$F\,\mathrm{score} = \frac{\mathrm{hits}}{\mathrm{hits} + \frac{1}{2}(\mathrm{misses} + \mathrm{false\ alarms})}. \tag{4}$$

Grid-based verification is an important, but not comprehensive, verification strategy (Wolff et al. 2014). This is particularly true for finer-scale (and often highly impactful) meteorological features as the metrics do not award skill for forecasts that are spatiotemporally close to, but not overlapping with, observations (Davis et al. 2006a,b). One option to address this concern is "fuzzy" verification, which loosens requirements to allow for some offset between forecasts and observations in space, time, or intensity (Ebert 2008). When the loosened restrictions occur in space, the metric is generally referred to as a neighborhood metric. The most common example of this is the fractions skill score (FSS; Roberts and Lean 2008). Like ETS, FSS binarizes the forecast and observation fields based on a specified threshold. However, FSS uses a neighborhood with a user-specified radius to determine a neighborhood average value for observations at a given pixel and compares those values for the forecast and observation fields. For further details, we refer the reader to Roberts and Lean (2008).

An alternative to neighborhood verification that also attempts to award skill for "closeness" is object-based verification (Davis et al. 2006a,b). Object-based verification matches contiguous forecasts and observed features based on intrinsic properties, such as area, distance, and shape. For example, Wolff et al. (2014), Yan and Gallus (2016), and Jensen et al. (2023) compare centroid distance, boundary distance, orientation angle difference, object–area ratio, and intersection divided by union area ratio, each with unique weighting. A scoring function uses these weights to calculate an "interest score," a measure of similarity between forecast and observed objects. Even if features do not overlap, similar values for these properties yield a higher interest score, representing greater forecast accuracy. The benefits of this strategy are clearly visualized in Fig. 1 of Davis et al. (2006a). A shortcoming of object-based verification is the properties and weights that compose the interest score are user defined and generally need to be determined empirically based on the meteorological feature being evaluated. Duda and Turner (2024) showed that results can be quite sensitive to these choices.

In this study, we provide examples of grid-based, neighborhood, and object-based verification, similar to Wolff et al. (2014) and Yan and Gallus (2016). We first investigate GFS, IFS, $\mathrm{GRAP}_{\mathrm{GFS}}$, $\mathrm{GRAP}_{\mathrm{IFS}}$, ERA5, and stage IV quantiles and frequency distributions at accumulation thresholds of 0.01, 0.05, 0.10, 0.25, 0.50, 0.75, and 1.00 in. across different forecast leads. GFS, $\mathrm{GRAP}_{\mathrm{GFS}}$, and $\mathrm{GRAP}_{\mathrm{IFS}}$ 6-hourly accumulated

precipitation forecasts are then verified against ERA5 and stage IV using continuous grid-based metrics like RMSE and bias at forecast leads of 12, 24, 48, 72, 96, 120, 144, and 168 h to capture any temporal trends in performance. While RMSE is aggregated across time and space, we show bias aggregated in time but not space to depict any consistent spatial signals.

We then extend to grid-based contingency table metrics, such as SEEPS and ETS. We used the dry probability and light/heavy threshold climatology produced by Rasp et al. (2024). The dry threshold was set to 0.01 in. (~0.25 mm), and we also used the scoring matrix presented in (Rasp et al. 2024). We display (1 − SEEPS) so that higher scores represent more skillful forecasts. We then investigate ETS with the same six-hour precipitation accumulation thresholds as the frequency distribution analysis to determine how each model performs on extremely light precipitation through intense precipitation.

Moving beyond grid-based verification, we calculate FSS, a neighborhood metric. FSS requires specifying both a threshold to binarize the field and a spatial radius (in grid cells) over which to award forecast skill. We tested thresholds of 0.01, 0.25, 0.50, and 1.00 in. and radii of 1, 2, 3, 5, and 10 grid cells. The Pysteps (Pulkkinen et al. 2019) package was used to perform this calculation.

Finally, we perform object-based verification. This is accomplished using the Method for Object-Based Diagnostic Evaluation (MODE; Davis et al. 2006a,b) in the Developmental Testbed Center's Model Evaluation Toolkit (MET). Following the precedent of Wolff et al. (2014), Yan and Gallus (2016), and Jensen et al. (2023), we compare centroid distance (weight = 2), boundary distance (weight = 4), orientation angle difference (weight = 1), object–area ratio (weight = 1), and intersection divided by union area ratio (weight = 2) to calculate the interest scores. We do this for forecast leads of 1, 3, and 5 days and at accumulation thresholds of 0.01, 0.25, and 0.50 in. These comparisons are done between each forecast and observation dataset, all on the 0.25° grid. These results should be taken with some caution, as recent work has shown that results can be sensitive to these choices (Duda and Turner 2024), and though we follow MODE precedent for precipitation evaluation, we do not perform a sensitivity analysis.

As Davis et al. (2009) show, interest scores can then be used to find the median of maximum interest (MMI). We refer the reader to Davis et al. (2009) for details on this metric but note that it represents an average of the similarity between forecast and observed features. MMIs can be compared across forecast systems, with higher MMIs indicating better matching features. Other metrics calculated using the full distribution of forecast and observed features include the total area ratio and the matched-to-total area ratio. The total area ratio is the summation of all the forecast feature areas divided by the summation of all observed feature areas and is representative of potential area biases [Eq. (5); Davis et al. 2009]. The matched-to-total area ratio is the summation of areas of all forecast and observed features that were matched to a corresponding feature compared to the total areas of all forecast and observed features [Eq. (6); Davis et al. 2009]. In a perfect system, all forecast (observed) features would be matched to an observed (forecast) feature and result in a ratio of 1. Thus, the matched-to-total area ratio is a measure of forecast accuracy:

$$\text{total area ratio} = \frac{\sum \text{forecast areas}}{\sum \text{observed areas}}, \tag{5}$$

$$\text{matched to total area ratio} = \frac{\sum \text{matched forecast areas} + \sum \text{matched observed areas}}{\sum \text{forecast areas} + \sum \text{observed areas}}. \tag{6}$$

One can also investigate specific properties of matched features to determine intensity biases (Davis et al. 2009). For each forecast–observation feature pair, we calculate the difference between the 90th percentile of the forecast feature intensity distribution and the 90th percentile of the observed feature intensity distribution (Davis et al. 2009). These differences are then summed across all feature pairs and divided by the number of feature pairs to find the mean 90% intensity difference as shown in Eq. (7). This metric is a representation of bias at the upper end of the precipitation distribution. We use a similar procedure but find the intensity sum (or total precipitation integrated across the area of a given precipitation object) of the observed feature intensity distribution rather than the 90th percentile for the mean summed intensity difference [Eq. (8)]. This is representative of bias in total area-integrated precipitation:

$$\text{mean 90\% intensity difference} = \frac{\sum(\text{forecast intensity}_{90} - \text{observed intensity}_{90})}{\text{matched feature count}}, \tag{7}$$

$$\text{mean intensity sum} = \frac{\sum(\text{forecast intensity}_{\text{sum}} - \text{observed intensity}_{\text{sum}})}{\text{matched feature count}}. \tag{8}$$

3. Results

a. Distributions

Quantile–quantile (Q–Q) plots for forecast precipitation distributions against Fig. 1a ERA5 and Fig. 1b stage IV precipitation distributions at a 72-h forecast lead are shown in Fig. 1. A forecast distribution equivalent to the reference distribution would fall on the 45° 1:1 line. The GFS and IFS exhibit similar behaviors, closely following the ERA5 1:1 line until about 0.50 in., after which both tend to overestimate the intensity of higher quantiles. Compared to stage IV, both the GFS and the IFS somewhat underestimate the intensity of higher quantiles. GRAP_{GFS} and GRAP_{IFS} show greater divergence from the observed distributions, greatly underestimating the intensity of higher quantiles. GRAP_{GFS} also slightly overestimates the intensity of lower quantiles, a behavior that is less prominent in GRAP_{IFS}. Though we only show the Q–Q plot for a 72-h forecast lead, other forecast leads yield similar results.

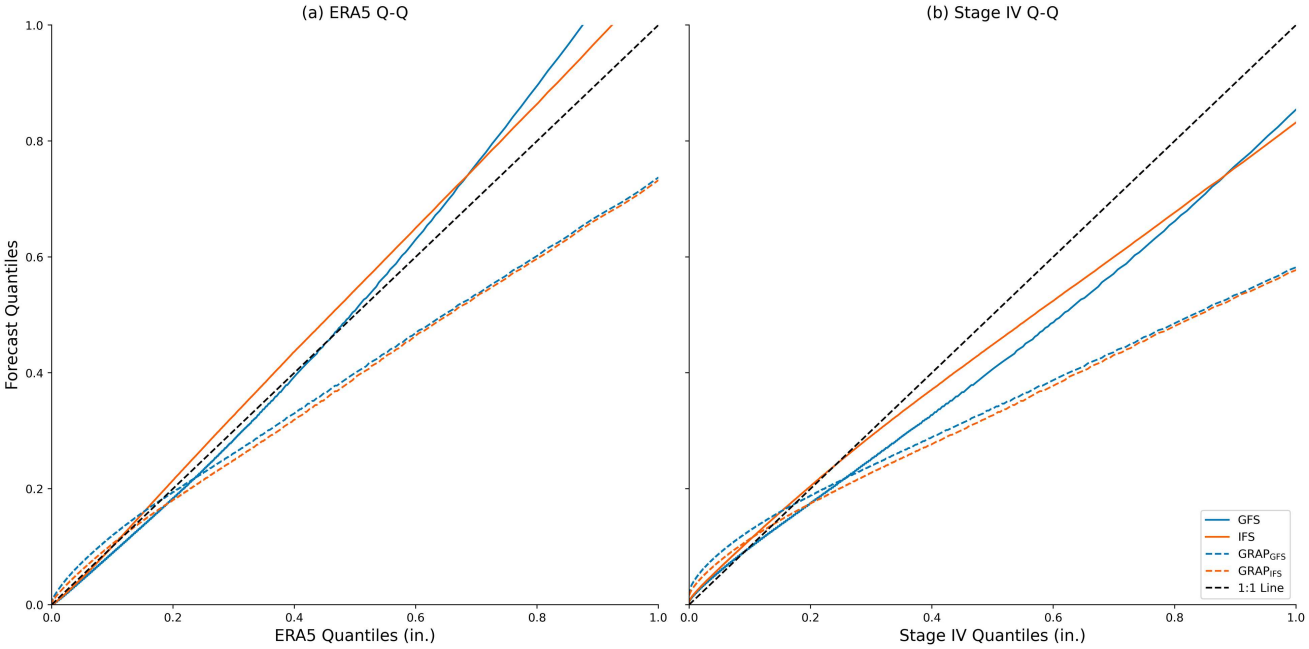


FIG. 1. Q–Q plots for the distributions of each forecast dataset compared to the (a) ERA5 distribution and (b) stage IV distribution. Blue solid (dashed) lines correspond to the GFS (GRAP$_{GFS}$), orange solid (dashed) lines correspond to the IFS (GRAP$_{IFS}$), and the black dashed line corresponds to the 1:1 line.

Ideally, the precipitation distributions should be nearly constant with the forecast lead. We can see if this is the case by stratifying by precipitation accumulation and plotting over time (Fig. 2). As expected, the ERA5 and stage IV distributions are constant over time. The GFS and IFS frequencies demonstrate some variation, but without any clear upward or downward trends. GRAP$_{IFS}$ has a slight upward trend at the 0.01-in. threshold and slight downward trends at the 0.25-, 0.50-, and 1.00-in. thresholds. GRAP$_{GFS}$ also has slight downward trends at the 0.25- and 0.50-in. thresholds but is largely constant at the 0.01-in. threshold and shows a slight upward trend at the 1.00-in. threshold. Reiterating what was found in Fig. 1, the GFS and IFS frequencies generally fall in between the ERA5 and stage IV frequencies, save for lower GFS frequencies at the 0.25-in threshold. GRAP$_{GFS}$ and GRAP$_{IFS}$ greatly overestimate the 0.01-in. frequencies compared to both the ERA5 and stage IV and underestimate frequencies of all other thresholds.

b. Grid-based metrics

1) RMSE AND SEEPS

We begin this section by calculating the two precipitation metrics already presented in Lam et al. (2023) supplemental material: RMSE (Fig. 3) and SEEPS (Fig. 4). Our RMSE results largely affirm those presented in Lam et al. (2023) supplemental material. Figure 3a shows GRAP$_{GFS}$ RMSE is about 0.02 in. lower than the GFS when comparing against ERA5, while GRAP$_{IFS}$ RMSE is similarly lower than the IFS. These results hold regardless of the forecast lead. GRAP$_{GFS}$ and GRAP$_{IFS}$ also have lower RMSE than their NWP counterparts when compared to stage IV, though to a slightly lesser magnitude than verified against ERA5. SEEPS results when each model is verified against ERA5 and stage IV are shown in Fig. 4. Again, our results are similar to Lam et al. (2023), with GRAP$_{GFS}$ and GRAP$_{IFS}$ receiving higher (1–SEEPS) values at all forecast leads than the GFS and IFS verified against both the ERA5 and stage IV observational datasets. The performance advantages of both GraphCast versions are slightly reduced for stage IV compared to ERA5. The error reduction in GRAP$_{GFS}$ compared to the GFS is slightly smaller than the error reduction in GRAP$_{IFS}$ compared to the IFS.

2) SPATIAL BIASES

In Fig. 5, we show the temporally averaged bias across the United States at a forecast lead of 72 h. There is little variation in these biases based on forecast lead. Perhaps the most notable feature is a GFS dry bias across the southeastern United States and Florida, in particular, when compared to both the ERA5 and stage IV datasets. The IFS also shows slight dry biases in Florida, though to a much lesser degree. In contrast, GRAP$_{GFS}$ and GRAP$_{IFS}$ have small biases in this region. This is a promising result for GraphCast, and further investigation of the specific circumstances (perhaps tropical activity or convection) leading to improvement in this region is warranted. Outside of the southeast, there are large differences between GRAP$_{GFS}$ and GRAP$_{IFS}$. While GRAP$_{IFS}$ is for the most part unbiased, GRAP$_{GFS}$ shows a small but widespread wet bias across the central United States and Mountain West. This is consistent with the greater frequency of 0.01-in. accumulations for GRAP$_{GFS}$ shown in Figs. 1 and 2 and could be a consequence of the unfamiliar GFS initial conditions.

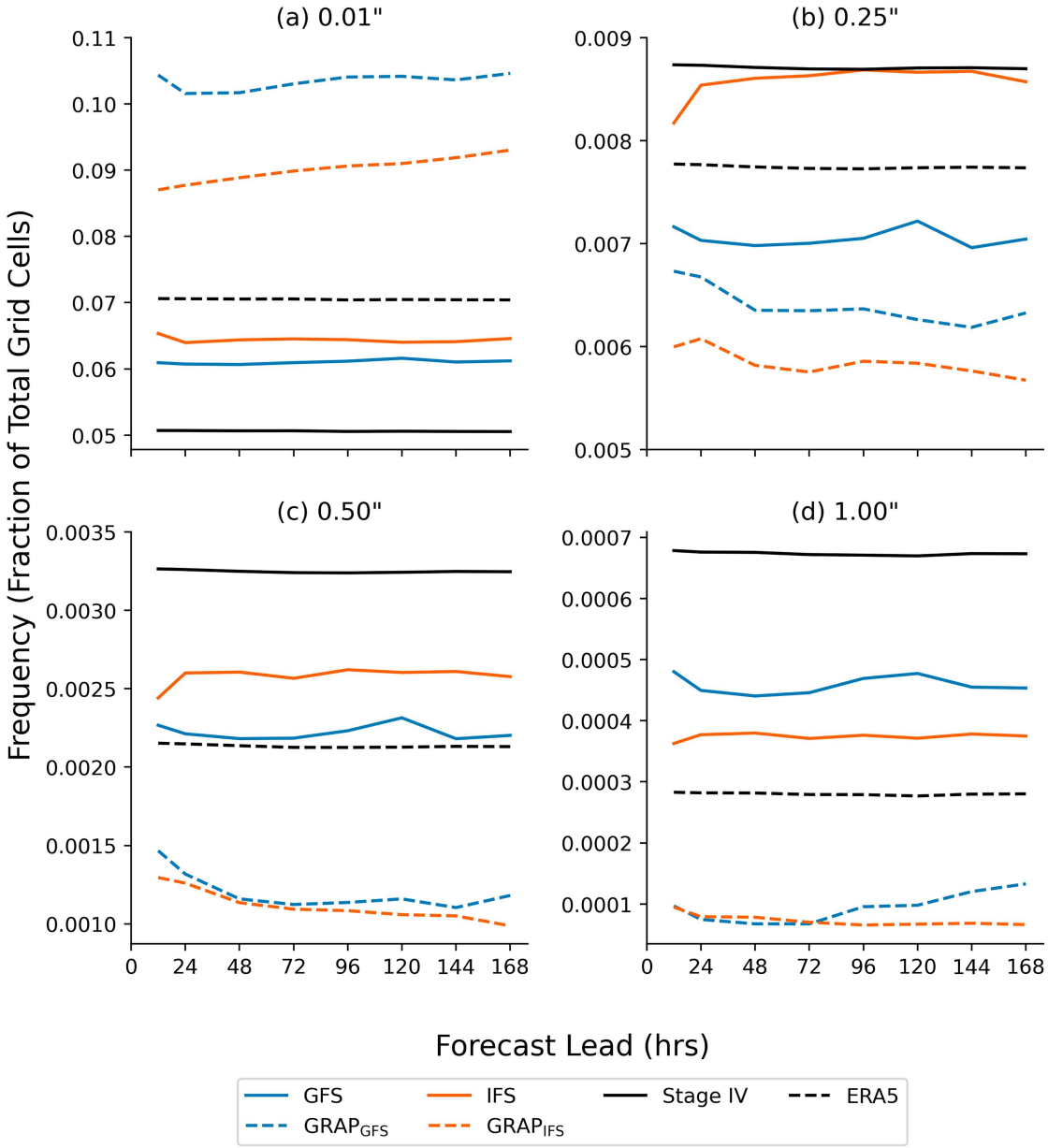


FIG. 2. Frequencies as a fraction of total grid cells of (a) 0.01, (b) 0.25, (c) 0.50, and (d) 1.00 in. by forecast lead. Blue solid (dashed) lines correspond to GFS (GRAP$_{GFS}$), and orange solid (dashed) lines correspond to IFS (GRAP$_{IFS}$). Solid black lines represent stage IV, and black dashed lines represent ERA5.

3) EQUITABLE THREAT SCORE

We next calculate the ETS for each forecast model and observational dataset pair. The results for forecasts at 1-, 3-, 5-, and 7-day leads are shown in Fig. 6. GRAP$_{GFS}$ outperforms the GFS and GRAP$_{IFS}$ outperforms the IFS at all intensity thresholds and all forecast leads when they are compared against ERA5, with the greatest advantage at low to moderate precipitation thresholds. In general, the difference in skill at all thresholds diminishes at the longer forecast leads. Results for comparisons to stage IV are slightly more mixed. GRAP$_{GFS}$ generally still outperforms the GFS at low to moderate thresholds, but the GFS gains a slight advantage beyond a threshold of approximately 0.50 in. GRAP$_{IFS}$ and the IFS converge at around 0.75 in. Note, however, that none of these models are particularly skillful at these higher-intensity thresholds.

c. *Neighborhood metrics*

The generally higher RMSE for the NWP models combined with their precipitation distributions more closely matching ERA5 and stage IV implies that there is a greater opportunity for the NWP models to gain ground on neighborhood metrics, like the FSS, that award skill for small

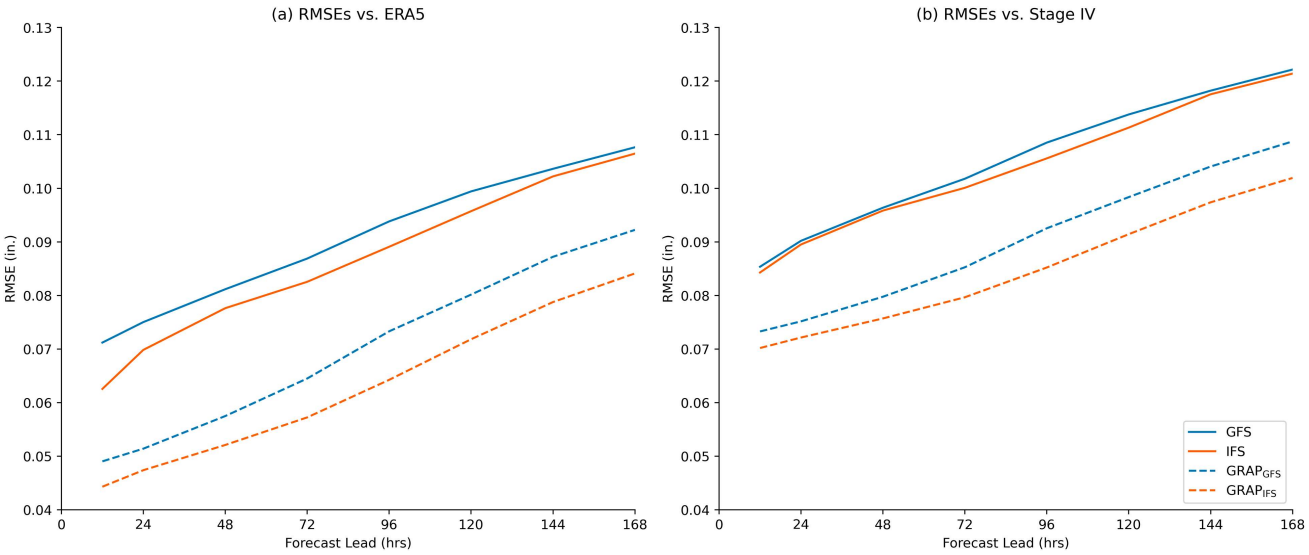


FIG. 3. RMSE by forecast lead for forecast datasets compared to (a) ERA5 and (b) stage IV. Blue solid (dashed) lines correspond to GFS (GRAP$_{GFS}$), and orange solid (dashed) lines correspond to IFS (GRAP$_{IFS}$).

spatial displacements. In other words, these two prior findings may suggest that the GFS and IFS error is primarily spatial error rather than intensity error. We can see this effect in Fig. 7, which shows FSS against neighborhood size for different accumulation thresholds at a 72-h forecast lead (other forecast leads demonstrate similar patterns) and against both ERA5 and stage IV. Verified against the ERA5, GRAP$_{GFS}$ and GRAP$_{IFS}$ receive higher scores at all but the 1-in. threshold for the smallest neighborhoods. However, the GFS and IFS gain ground on GRAP$_{GFS}$ and GRAP$_{IFS}$ at lower thresholds and surpass GraphCast at higher thresholds as the neighborhood size increases. These patterns are similar when verified against stage IV, though more dramatic, with the GFS and IFS quickly catching up to and surpassing GRAP$_{GFS}$ and GRAP$_{IFS}$ at all thresholds with increasing neighborhood size.

d. Object-based verification

1) MEDIAN MAXIMUM INTEREST

MMI scores for three different forecast leads and three different accumulation thresholds are shown for each model/observation pair in Fig. 8. The results are highly dependent on the verification dataset, forecast lead, and likely MODE configuration (Duda and Turner 2024; we did not perform a sensitivity analysis). Against ERA5, GRAP$_{GFS}$ and GRAP$_{IFS}$ generally have slightly higher MMI than the GFS and IFS at the 0.01- and 0.25-in. thresholds. At the 0.50-in. threshold, results are more mixed, with GRAP$_{GFS}$ and GRAP$_{IFS}$ performing better at a 24-h lead and GFS and IFS performing better at longer leads. Against stage IV, GRAP$_{GFS}$ and GRAP$_{IFS}$ perform comparably to the GFS and IFS at the 0.01-in. threshold, slightly worse at the 0.25-in. threshold, and worse at the 0.50-in. threshold.

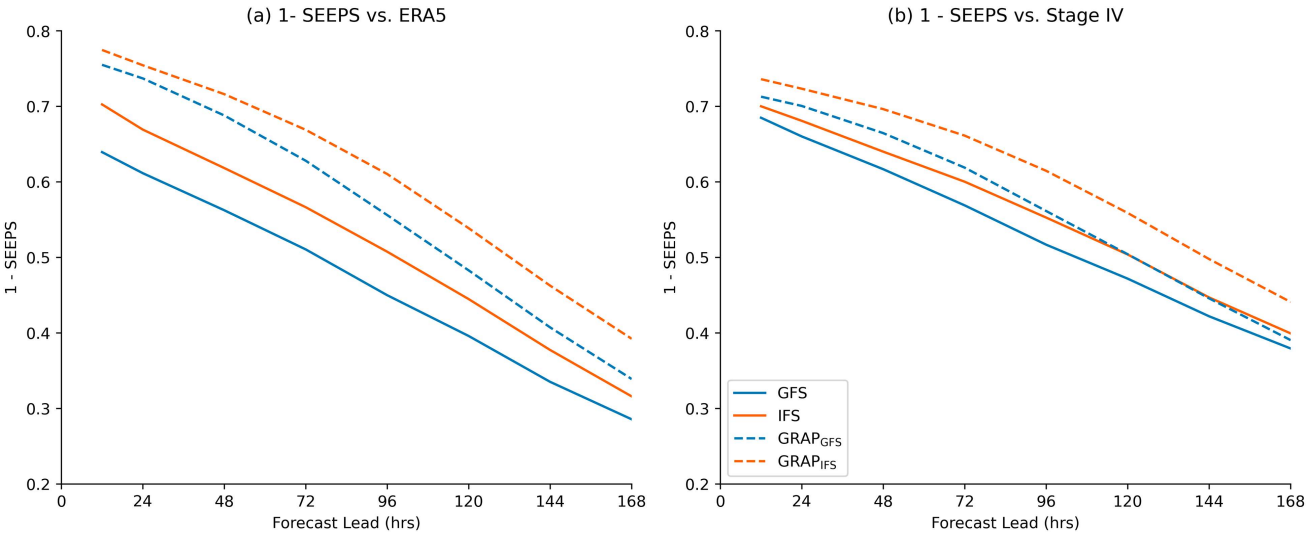


FIG. 4. (1 − SEEPS) by forecast lead for forecast datasets compared to (a) ERA5 and (b) stage IV. Blue solid (dashed) lines correspond to GFS (GRAP$_{GFS}$), and orange solid (dashed) lines correspond to IFS (GRAP$_{IFS}$).

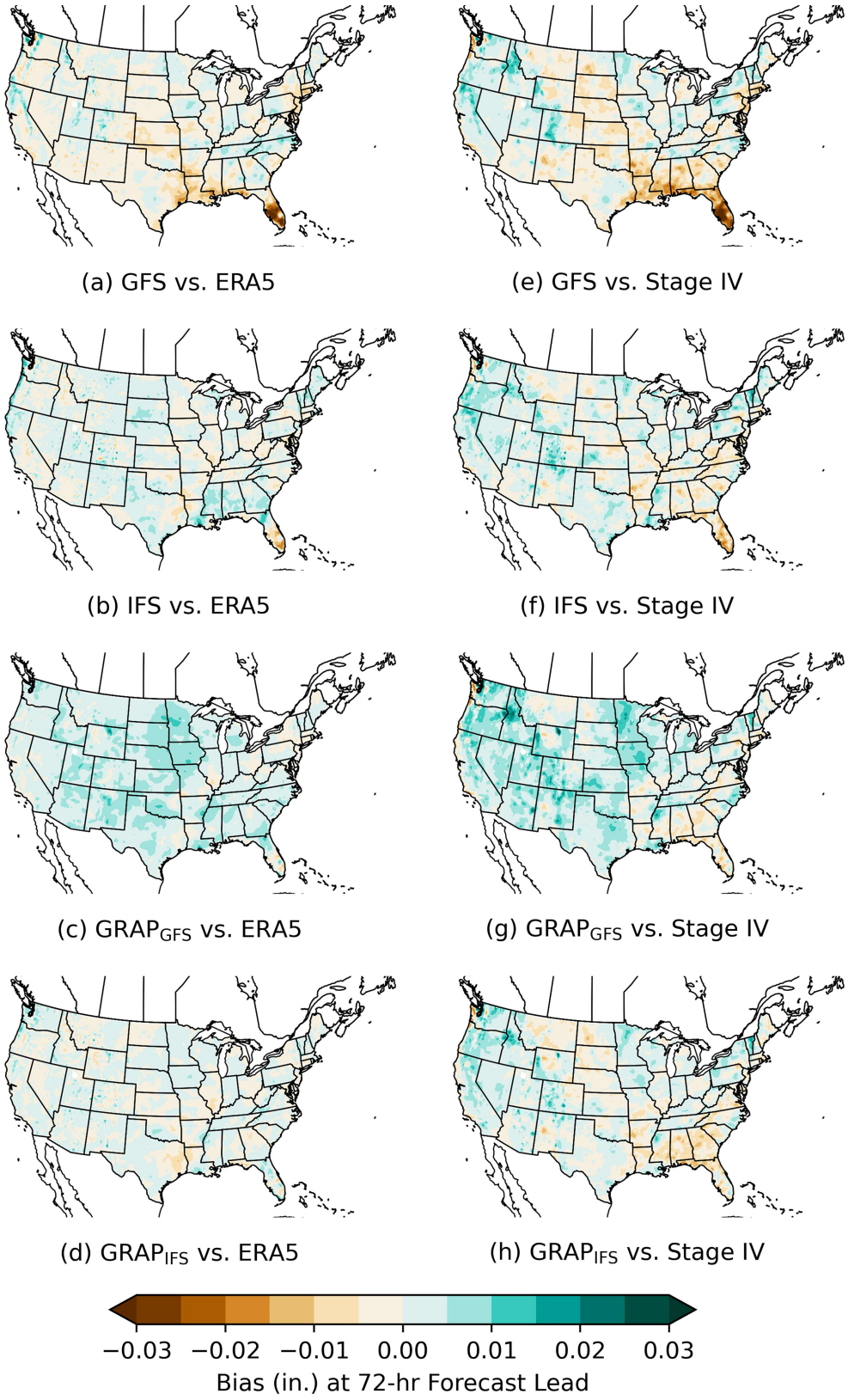


FIG. 5. 6-h precipitation accumulation average biases at a 72-h forecast lead for (a) GFS versus ERA5, (b) IFS versus ERA5, (c) GRAP$_{GFS}$ versus ERA5, (d) GRAP$_{GFS}$ versus ERA5, (e) GFS versus stage IV, (f) IFS versus stage IV, (g) GRAP$_{GFS}$ versus stage IV, (h) and GRAP$_{IFS}$ versus stage IV.

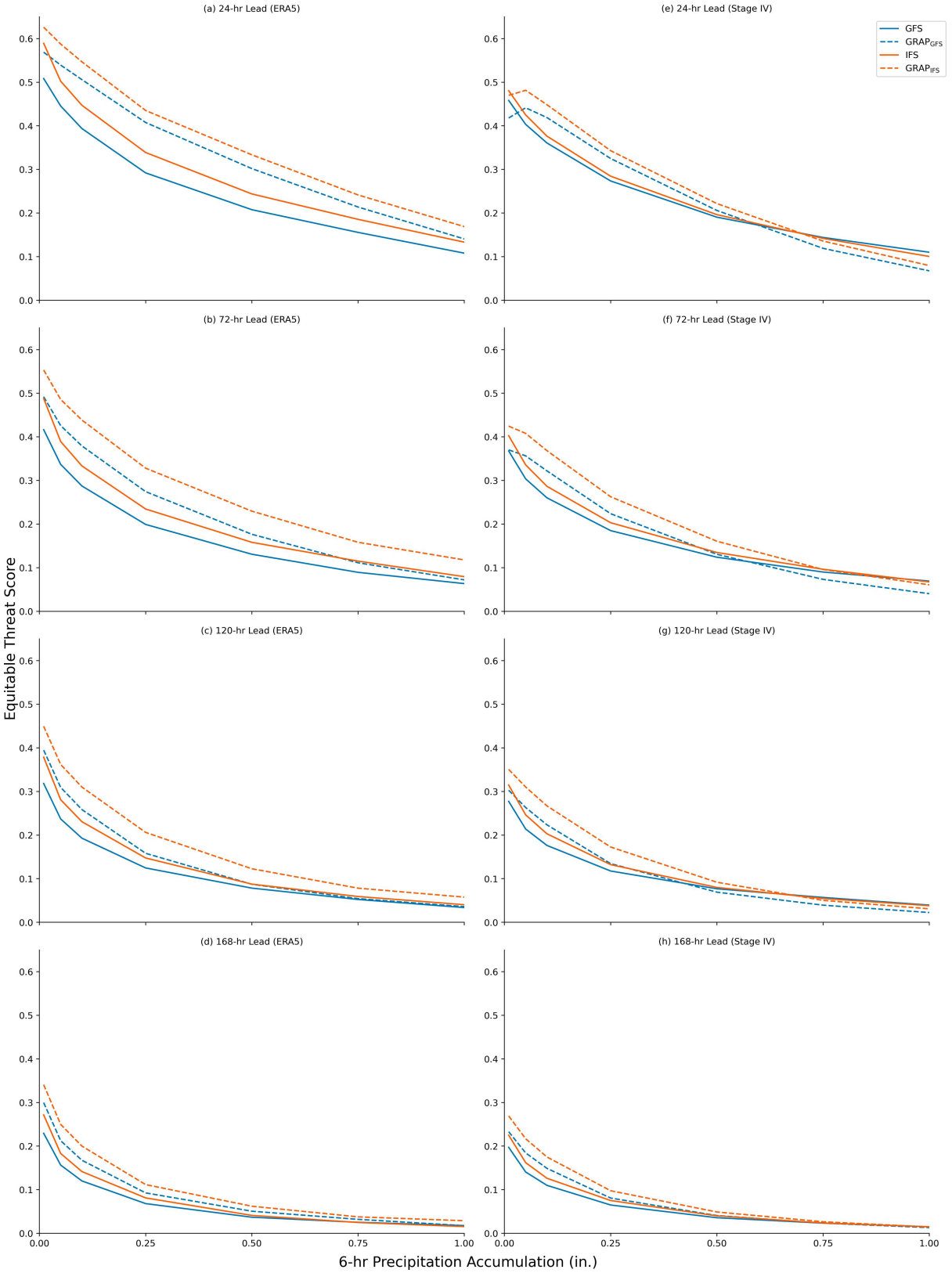


FIG. 6. ETS by 6-h precipitation accumulation for all forecast models compared to (a) ERA5 at 24 h, (b) ERA5 at 72 h, (c) ERA5 at 120 h, (d) ERA5 at 168 h, (e) stage IV at 24 h, (f) stage IV at 72 h, (g) stage IV at 120 h, and (h) stage IV at 168 h. Blue solid (dashed) lines correspond to GFS ($GRAP_{GFS}$), and orange solid (dashed) lines correspond to IFS ($GRAP_{IFS}$).

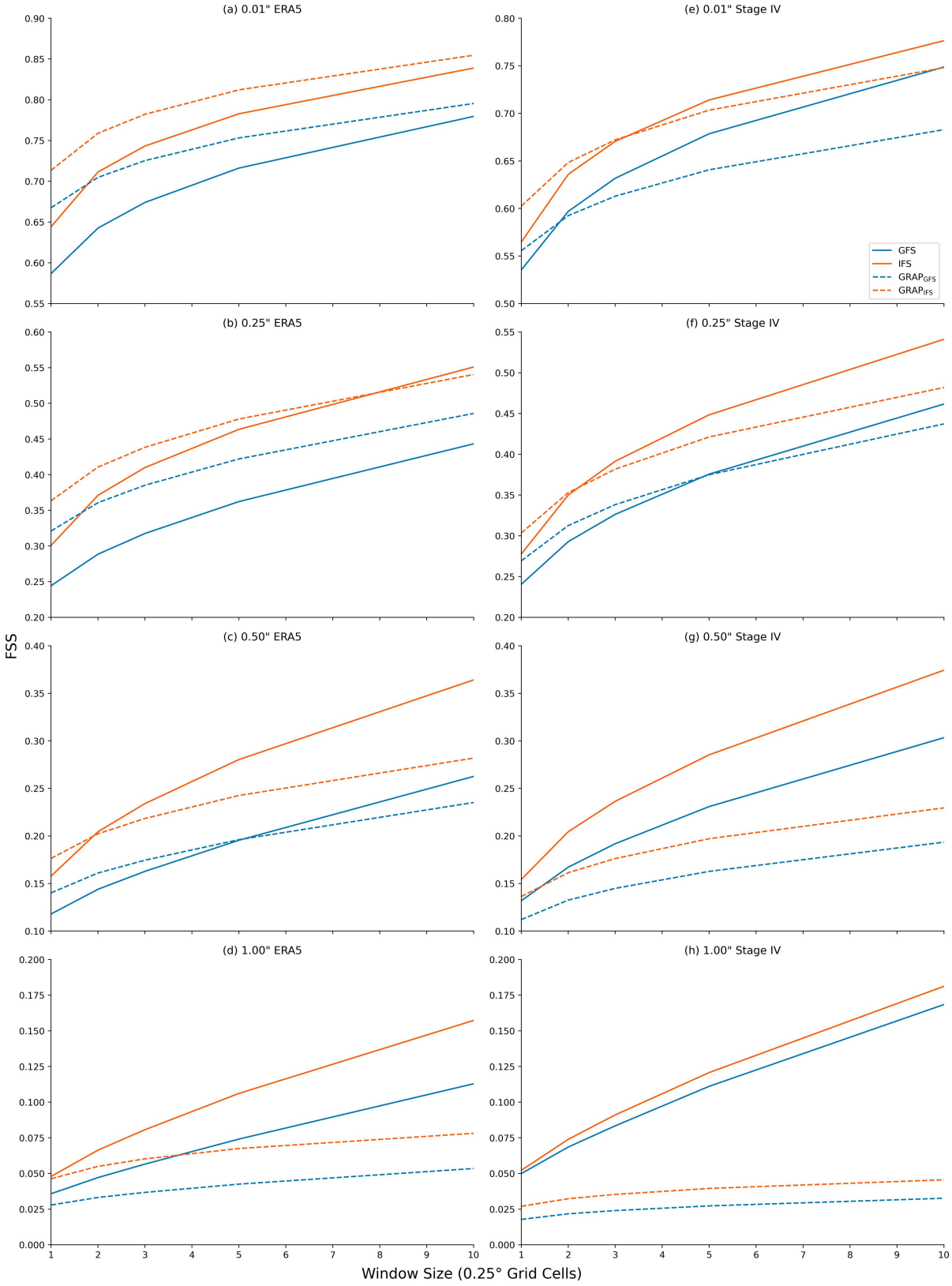


FIG. 7. FSS by neighborhood size (in 0.25° grid cells) for all forecast models compared to (a) ERA5 0.01 in., (b) ERA5 0.25 in., (c) ERA5 0.50 in., (d) ERA5 1.00 in., (e) stage IV 0.01 in., (f) stage IV 0.25 in., (g) stage IV 0.50 in., and (h) stage IV 1.00 in. at a 72-h lead. Blue solid (dashed) lines correspond to GFS ($GRAP_{GFS}$), and orange solid (dashed) lines correspond to IFS ($GRAP_{IFS}$).

	GFS-ERA5			GFS-Stage IV		
	24-hr	**72-hr**	**120-hr**	**24-hr**	**72-hr**	**120-hr**
0.01”	0.910	0.869	0.819	0.876	0.856	0.675
0.25”	0.845	0.541	0.245	0.839	0.522	0.250
0.50”	0.520	0.261	0.000	0.558	0.265	0.000
	IFS-ERA5			**IFS-Stage IV**		
0.01”	0.957	0.893	0.853	0.886	0.859	0.796
0.25”	0.880	0.688	0.470	0.857	0.601	0.425
0.50”	0.697	0.443	0.089	0.556	0.331	0.037
	GRAP$_{GFS}$-ERA5			**GRAP$_{GFS}$-Stage IV**		
0.01”	0.952	0.896	0.857	0.867	0.851	0.649
0.25”	0.887	0.690	0.196	0.826	0.470	0.013
0.50”	0.619	0.000	0.000	0.000	0.000	0.000
	GRAP$_{IFS}$-ERA5			**GRAP$_{IFS}$-Stage IV**		
0.01”	0.972	0.930	0.869	0.883	0.866	0.832
0.25”	0.898	0.841	0.453	0.838	0.507	0.231
0.50”	0.823	0.246	0.000	0.000	0.000	0.000

FIG. 8. MMI score for three different forecast leads and three different accumulation thresholds for each pair of forecast and observation datasets. Greater saturation corresponds to greater MMI (or a smaller difference from a perfect score of 1).

2) TOTAL AREA RATIO AND MATCHED-TO-TOTAL AREA RATIO

The total area ratio and the matched-to-total area ratio are shown for a forecast lead of 72 h in Fig. 9, but findings are again consistent at different forecast leads. A reminder is that the total area ratio represents the summed areas of all forecast features divided by the summed areas of all observed features, with values of one indicating equal total areas, values

	(a) Total Area Ratio							
	GFS - ERA5	IFS - ERA5	GR_{GFS} - ERA5	GR_{IFS} - ERA5	GFS - Stage IV	IFS - Stage IV	GR_{GFS} - Stage IV	GR_{IFS} - Stage IV
0.01"	0.934	0.969	1.317	1.159	1.150	1.193	1.622	1.426
0.25"	0.895	1.083	0.863	0.780	0.780	0.943	0.752	0.679
0.50"	1.043	1.020	0.603	0.584	0.723	0.707	0.418	0.405
	(b) Matched to Total Area Ratio							
	GFS - ERA5	IFS - ERA5	GR_{GFS} - ERA5	GR_{IFS} - ERA5	GFS - Stage IV	IFS - Stage IV	GR_{GFS} - Stage IV	GR_{IFS} - Stage IV
0.01"	0.835	0.860	0.750	0.785	0.837	0.826	0.740	0.807
0.25"	0.739	0.814	0.877	0.846	0.731	0.774	0.743	0.779
0.50"	0.622	0.705	0.690	0.748	0.612	0.657	0.590	0.639

FIG. 9. (a) Total area ratio between GFS, IFS, $GRAP_{GFS}$ (further abbreviated to GR_{GFS}), and $GRAP_{IFS}$ (further abbreviated to GR_{IFS}) forecast feature areas and ERA5 and stage IV observed feature areas at a 72-h forecast lead. Brown (green) hues represent a smaller (greater) area than observed. (b) Matched area ratio between GFS, IFS, $GRAP_{GFS}$, and $GRAP_{IFS}$ and ERA5 and stage IV as defined in Eq. (6). Greater saturation represents a greater fraction of matched area, with a value of 1 representing a perfect match.

greater than one indicating an overforecast of feature areas, and values less than one indicating an underforecast of feature areas. The matched-to-total area ratio represents the total area of precipitation features that are successfully matched to one another divided by the total area of all precipitation features. Ideally, all forecast features would be matched to observed features and this value would be one. No matching feature would result in a value of zero.

The IFS and GFS generally have total area ratios close to one against ERA5 at all thresholds. Compared to stage IV, they both have slight positive biases at the 0.01-in. threshold and negative area biases at the 0.50-in. threshold. $GRAP_{GFS}$ has a large positive area bias when compared to both ERA5 and stage IV at the 0.01-in. threshold. $GRAP_{IFS}$ also exhibits positive biases at the 0.01-in. threshold but to a lesser degree than $GRAP_{GFS}$. At the 0.25- and 0.50-in. thresholds, $GRAP_{GFS}$ and $GRAP_{IFS}$ both have negative biases. In other words, GraphCast-forecast precipitation features greater than 0.01 in. are larger than observed, while GraphCast-forecast precipitation features greater than 0.50 in. tend to be smaller than observed.

The matched-to-total area ratios have less consistent patterns. Verified against ERA5, both the GFS and ERA5 see a decrease in matched area proportion with increasing threshold. In contrast, $GRAP_{GFS}$ and $GRAP_{IFS}$ actually see an increase in matched area proportion from 0.01 to 0.25 in., followed by a decrease at 0.50 in. Verified against stage IV, the four models show similar proportions of matched area, with the exception of $GRAP_{GFS}$, which has a lower proportion of matched area at the 0.01-in. threshold. The lower proportions of matched areas for GraphCast at the 0.01-in. threshold are likely related to the greater frequencies of $GRAP_{GFS}$-forecast 0.01-in. precipitation (Figs. 1 and 2), as more of its low-intensity features go unmatched with observations.

3) MEAN 90% INTENSITY AND INTENSITY SUM DIFFERENCES

As in Davis et al. (2009) and Yan and Gallus (2016), we calculated the mean of the 90th-percentile precipitation intensity and intensity sum of matched features and report the results in Fig. 10. A reminder that the mean 90% intensity difference represents the pairwise difference between the 90th percentile of a forecast precipitation feature and the 90th percentile of a matched observed precipitation feature. Similarly, the mean intensity sum difference represents the pairwise difference between the total integrated precipitation over a forecast feature's area and the total integrated precipitation over a matched observed feature's area. Statistics are again shown for a forecast lead of 72 h, but findings are consistent for different forecast leads.

Compared to the ERA5, all four models are relatively unbiased for the 0.01-in. threshold for mean 90% intensity. At the 0.25- and 0.50-in. thresholds, the GFS and, to a lesser extent, the IFS show small high-intensity biases, while $GRAP_{GFS}$ and $GRAP_{IFS}$ have more substantial low-intensity biases. This means that the 90th percentile of intense precipitation features is somewhat overestimated by the GFS and IFS and is underestimated by GraphCast. The GFS features' 90th percentiles generally align with stage IV, while the IFS slightly underestimates and $GRAP_{GFS}$ and $GRAP_{IFS}$ substantially

	(a) Mean 90% Intensity Differences							
	GFS - ERA5	**IFS - ERA5**	**GR$_{GFS}$ - ERA5**	**GR$_{IFS}$ - ERA5**	**GFS - Stage IV**	**IFS - Stage IV**	**GR$_{GFS}$ - Stage IV**	**GR$_{IFS}$ - Stage IV**
0.01"	0.001	0.013	-0.024	-0.028	-0.041	-0.025	-0.062	-0.067
0.25"	0.101	0.032	-0.202	-0.194	-0.060	-0.136	-0.369	-0.358
0.50"	0.177	0.087	-0.313	-0.302	-0.007	-0.163	-0.537	-0.518
	(b) Mean Intensity Sum Differences							
	GFS - ERA5	**IFS - ERA5**	**GR$_{GFS}$ - ERA5**	**GR$_{IFS}$ - ERA5**	**GFS - Stage IV**	**IFS - Stage IV**	**GR$_{GFS}$ - Stage IV**	**GR$_{IFS}$ - Stage IV**
0.01"	-5.959	-1.732	6.967	1.477	-4.888	1.489	12.963	5.379
0.25"	-10.204	-5.609	-10.028	-12.917	-21.096	-15.995	-19.983	-22.055
0.50"	3.755	-6.010	-25.336	-27.126	-18.340	-30.218	-52.521	-54.643

FIG. 10. (a) Differences between GFS, IFS, GRAP$_{GFS}$ (further abbreviated to GR$_{GFS}$), and GRAP$_{IFS}$ (further abbreviated to GR$_{GFS}$) forecast mean 90th-percentile intensity and ERA5 and stage IV observed mean 90th-percentile intensity at a 72-h forecast lead. (b) Differences between GFS, IFS, GRAP$_{GFS}$, and GRAP$_{IFS}$ forecast mean intensity sum and ERA5 and stage IV observed mean intensity sum at a 72-h forecast lead. Positive (wet) biases are shaded in green and negative (dry) biases are shaded in brown with greater saturation representing a greater magnitude of bias.

underestimate the 90th percentiles of 0.25- and 0.50-in. features.

Intensity sum differences at the 0.01-in. threshold are generally small, with the GFS exhibiting modest dry biases and GRAP$_{GFS}$ exhibiting modest wet biases. All models have comparable dry biases at the 0.25-in. threshold, though the IFS has a small advantage. Finally, at the 0.50-in. threshold, GRAP$_{GFS}$ and GRAP$_{IFS}$ have large dry biases compared to either verification dataset. The IFS also has dry biases but is smaller than GraphCast. The GFS has an even smaller dry bias when compared to stage IV and a small wet bias compared to ERA5. To summarize, results are somewhat mixed at the 0.01-in. threshold, all models have a similar dry bias at the 0.25-in. threshold, and GRAP$_{GFS}$ and GRAP$_{IFS}$ have large dry biases at the 0.50-in. threshold.

4. Discussion

We performed an independent, comparative precipitation verification of the GFS and IFS with GFS- and IFS-initialized versions of GraphCast over the contiguous United States. We presented multiple types of verification, including grid based, neighborhood, and object oriented, and verified against both ERA5 and stage IV datasets.

GRAP$_{GFS}$ and GRAP$_{IFS}$ held advantages over the GFS and IFS for most grid-based metrics. In particular, RMSE and SEEPS were both substantially lower for the two versions of GraphCast compared to their NWP counterparts, affirming the results of Lam et al. (2023). This is true when verifying with either ERA5 or stage IV data, though GraphCast has a somewhat greater advantage when verified with ERA5. GRAP$_{GFS}$ and GRAP$_{IFS}$ ETSs were also higher than the GFS and IFS for all forecast leads and all thresholds when verified with ERA5. Verified with stage IV, GRAP$_{GFS}$ and GRAP$_{IFS}$ generally still have an advantage at light to moderate thresholds, while the GFS and IFS catch up at more intense precipitation thresholds.

One potential drawback to GRAP$_{GFS}$ and GRAP$_{IFS}$ lies in their frequency distributions. In particular, both versions of GraphCast tend to overestimate the frequency of light precipitation and underestimate the frequency of intense precipitation compared to either ERA5 or stage IV. The GFS and IFS generally have distributions falling somewhere in between ERA5 and stage IV.

At the smallest neighborhood sizes, GRAP$_{GFS}$ and GRAP$_{IFS}$ have higher FSS than the GFS and IFS. However, as the neighborhood size increases, the GFS and IFS gain ground on, and in some cases surpass, GRAP$_{GFS}$ and GRAP$_{IFS}$. This is seen in verification against ERA5 but is more apparent against stage IV. It is also most evident at higher accumulation thresholds. This likely suggests that GraphCast's advantage is primarily driven by lower spatial error rather than better intensity forecasts, as the advantage is reduced when spatial offsets are allowed.

Including both the native IFS-initialized version and a previously untested GFS-initialized version of GraphCast allows us to make some conclusions about the validity of swapping initial condition sets. GRAP$_{IFS}$ consistently outperforms

$GRAP_{GFS}$ by a small margin, but similar magnitudes of performance gains are made on grid-based metrics between the IFS and $GRAP_{IFS}$ and the GFS and $GRAP_{GFS}$. However, there appear to be some anomalies introduced with using GFS initial conditions. Namely, $GRAP_{GFS}$ more frequently makes light precipitation forecasts than $GRAP_{IFS}$, likely contributing to the widespread wet biases seen in Fig. 5. This behavior could perhaps be alleviated by fine-tuning GraphCast specifically on GFS data.

For the most part, similar conclusions can be made based on object-based verification using the MODE tool. $GRAP_{GFS}$ and $GRAP_{IFS}$ MMIs were higher than those of the GFS and IFS at 0.01- and 0.25-in. thresholds when verified against ERA5. $GRAP_{GFS}$ and $GRAP_{IFS}$ MMIs were comparable to GFS and IFS at the 0.01-in. threshold and lower at the 0.25- and 0.50-in. thresholds when verified against stage IV. The total area ratio showed large positive area biases for both versions of GraphCast at low-intensity thresholds and large negative area biases at high-intensity thresholds compared against either verification dataset. The area-integrated precipitation across GraphCast-forecast features showed slight wet biases for 0.01-in. objects and dry biases for 0.50-in. objects.

These findings also have potential operational applications. For example, the RMSE and neighborhood results suggest a potential forecast strategy of using GraphCast for forecasting the most likely position of precipitation features and the GFS for determining the potential of intense precipitation within the features. Due to the recency of AIWP precipitation forecasts and lingering questions about the capability of GraphCast to capture extreme precipitation, we suggest using AI-based models in tandem with physics-based models rather than choosing one over the other at this point.

The intent of this paper is not to make a conclusive judgment on whether GraphCast or GFS precipitation forecasts are better or to suggest that AI verification to this point is misleading. On the contrary, we are optimistic about the opportunities presented by pure AI models. Despite the fact that Lam et al. (2023) suggest that little effort was made to improve GraphCast precipitation forecasts, it already demonstrates comparable or better performance than NWP in many regards with many potential avenues for further improvements. We merely wished to explore the potential strengths and weaknesses of AIWP precipitation forecasts and some of the contextual factors important in verification. We hope that these results can serve as a baseline for comparison when inevitable new advancements are made in AIWP precipitation forecasts.

The key findings are summarized below:

1) We affirmed that GraphCast has lower RMSE and higher SEEPS than NWP for 6-h precipitation forecasts. ETS generally also favors GraphCast, but results are less consistent when verified with stage IV and at higher accumulation thresholds.
2) GraphCast overestimates the frequency of light precipitation accumulations and underestimates the frequency of higher accumulation thresholds.
3) GFS and IFS FSSs increase more than $GRAP_{GFS}$ and $GRAP_{IFS}$ with increasing neighborhood sizes, likely due to a more representative intensity distribution but with larger spatial error.
4) Initializing GraphCast with GFS initial conditions sees comparable performance gains to the IFS-initialized version, but some caution should be exercised as behavioral differences were noticed, such as a greater prevalence of light precipitation.

There is ample opportunity for future work in the field of AIWP precipitation forecasting. We noted that fine-tuning GraphCast on the GFS could further improve $GRAP_{GFS}$ performance. Applying alternate normalization and loss functions tailored for precipitation would also likely be beneficial. Future work is also not limited to model development and verification. Even if these models were to outperform physics-based NWP in every skill metric, we still need to assess forecaster perspectives of AIWP to determine the feasibility of transitioning these models from research to operations. Do forecasters view these models as trustworthy? What is their utility given their current spatiotemporal resolution and limited output variables? How do they fit into existing forecast processes? These are questions that may be less relevant to private sector development but have large implications for widespread adoption and operational implementation.

Acknowledgments. CIRA author Radford is supported by funding from NOAA Award NA19OAR4320073. CIRA author Ebert-Uphoff's work is supported by CIRA's ML Strategic Investment. The authors would like to compliment the developers of FourCastNet, Pangu-Weather, and GraphCast for publishing their model code. Open access code is of paramount importance toward advancing the AIWP field.

Data availability statement. The authors used a local archive of 0.25° GFS initial conditions. Those data are also available via Amazon Web Services (AWS; https://registry.opendata.aws/noaa-gfs-bdp-pds/). IFS initial conditions were attained with a research license granted by the ECMWF. Stage IV precipitation analysis data are available through NCAR (https://data.eol.ucar.edu/dataset/21.093). The ERA5 data (Hersbach 2020) were downloaded from the Copernicus Climate Change Service (C3S) (2023). The operational GraphCast model code used in this study is available on GitHub (https://github.com/google-deepmind/graphcast).

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